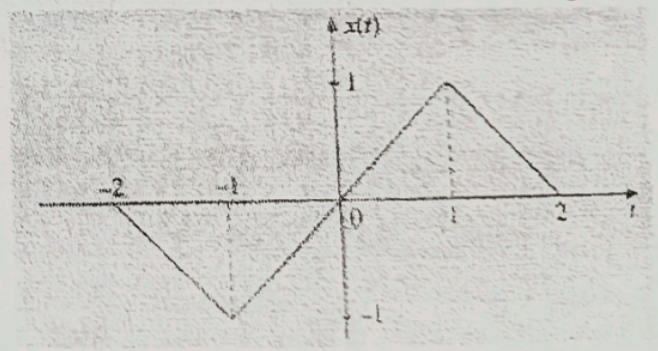
राष्ट्रीय प्रौद्योगिकी संस्थान, कुरुक्षेत्र

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

DEPARTMENT OF ELECTRICAL ENGINEERING

Date: 12-11-2025 **Signals and Systems (RAPC204), Mid Term Exam-II** **Time: 50 Min.**

All questions are compulsory **Total: 15 Marks**

1. Using Fourier transform, Evaluate the convolution of the signals $x(t) = te^{-t}u(t)$ and $h(t) = te^{-2t}u(t)$? [4 M]
2. Find the Laplace transform of $x(t) = \frac{\sin^2 at}{t}$ $u(t)$ [3 M]
3. Obtain the Fourier transform of the signal $x(t) = e^{-|t|}\sin 5|t|$ for all t? [4 M]
4. [4 M]

   Find the Fourier transform of the signal $x(t)$ shown in the figure using differentiation property of Fourier transform?